Pre-Class Assignment: Guest Speaker — Mary Reagan, PhD

Head of Product, Oumi AI (prev. DataRobot, Fiddler AI)

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

Guest speaker: Mary Reagan, PhD. Head of Product at Oumi AI (since March 2026; joined Oumi as Lead Product Manager in January 2026). Previously Senior Product Manager at DataRobot (2023–2025), working on end-to-end machine learning and generative-AI solutions. Before that, Data Scientist II → Product Manager → Senior PM at Fiddler AI (2020–2023), where she worked on, in her own words, *"explainable AI, fairness in machine learning, and monitoring of ML models."* She came to industry through DataKind, a non-profit that applies data science to humanitarian problems. Her PhD is from **Stanford in mineral physics** (Extreme Environments Lab, 2014–2019) — a reminder that the most interesting people in AI often arrive from outside computer science. She has published on bias and fairness in machine-learning systems, including the *Towards Data Science* article *"Understanding Bias and Fairness in AI Systems"* (2021).

Reading / preparation:

1. Skim Fiddler AI's homepage and product description at fiddler.ai (focus on ML/AI observability, guardrails, governance).
2. Skim DataRobot's homepage at datarobot.com (focus on the AI/agent platform, AutoML/MLOps, governance).
3. Skim Oumi's homepage at oumi.ai and its company page (focus on open-source fine-tuning of foundation models, custom models from prompts, the build/evaluate/synthesize/deploy loop).
4. Read Mary Reagan, *"Understanding Bias and Fairness in AI Systems,"* Towards Data Science (2021). Available at: medium.com/data-science/understanding-bias-and-fairness-in-ai-systems-6f7fbfe267f3.

Instructions: Answer each question in 1–3 sentences. These questions are designed to be straightforward if you did the preparation. You must submit this assignment in person at the start of class. The assignment must be handwritten. No photocopies will be accepted.

Why we do this: Dr. Reagan has built products at three companies that sit at three different layers of the AI value chain — model *monitoring* (Fiddler), enterprise AI *platforms* (DataRobot), and open-source foundation-model *training* (Oumi). She has also done research on bias and fairness, the topic of our Chapter 20 lecture. The point of this PCA is to make sure you can ask her substantive questions during the Q&A.

Part A — The three companies

1. **Fiddler AI.** In one or two sentences, what problem does Fiddler solve for a customer who has *already* deployed an AI model in production?
2. **DataRobot.** DataRobot describes itself as a platform that helps enterprises "build, operate, and govern" AI agents at scale. In one sentence each, what does *each* of those three verbs (build, operate, govern) refer to?
3. **Oumi.** Oumi's pitch is that a customer can describe what they want a model to do in plain language and get a working custom model in hours rather than weeks. In one or two sentences, what does Oumi do that a customer using a frontier model (e.g., GPT-class) directly would *not* get?

Part B — The economics behind these products (connect to what we covered)

4. **Chapter 5 (Complements to Prediction).** In Chapter 5, Gans argued that prediction and **judgment** are typically complements: a better prediction is more valuable when paired with better human judgment. Of Fiddler, DataRobot, and Oumi, which one most directly sells tools for the *judgment* step rather than the *prediction* step? Justify in one sentence.
5. **microGPT and Oumi.** In Lab 1, you retrained microGPT on dinosaur names. Oumi's product is, in spirit, a similar operation (fine-tune a pretrained model on a custom dataset) but not quite the same, and at production scale. In one or two sentences, what is Oumi doing that is the *same* as what you did in Lab 1, and what is it doing that Lab 1 did *not* do, and is there anything that you did in Lab 1 that Oumi does not do?

Part C — Bias and fairness (Dr. Reagan's own writing, and our Chapter 20)

6. In her Towards Data Science article, Dr. Reagan distinguishes several sources of bias in ML systems (e.g., historical bias, representation/sampling bias, measurement bias, algorithmic bias). Pick *any one* and, in one or two sentences, define it and give an example of where it would arise.
7. In our Chapter 20 reading, Rambachan et al. decompose group differences in AI predictions into base-rate, measurement-error, and estimation-error pieces. In one sentence, which of Dr. Reagan's bias categories from question 6 maps most cleanly onto Rambachan's **measurement-error** piece, and why?

Part D — One question to ask in person

8. Dr. Reagan has spent much of her career at companies that *sell tools to detect or reduce bias* (Fiddler, DataRobot) and that *let customers train their own models* (Oumi). Write one question that you would like to ask Dr. Reagan during the Q&A. The best questions will be specific to her experience rather than generic ("What's the future of AI?"). One or two sentences.